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Digitalization, trust, and sustainability transitions: Insights from two blockchain-based green experiments in China's electricity sector

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ABSTRACT

Linking decarbonization with digitalization has been a promising trend in transition research. This paper explores how blockchain technologies matter in sustainability transitions through their key mechanism—trust—by comparing two blockchain-based green experiments in China's electricity sector. Blockchain technologies can bring institution-based trust among participating agents, but other types of trust are needed to promote this institution-trust to a system level. Incumbent actors have a greater advantage in mobilizing established trust relationships to promote blockchain-based green experiments and strengthen their power positions. It implies that digitalization may create unequal opportunities for incumbents and newcomers, leading to imbalanced trust and power relationships.

1. Introduction

Digitalization and sustainability have become the two dominant 'social words' across the world (Lenz, 2021). There has been mounting research debating the impact of digitalization on sustainability or sustainable development goals (SDGs) given the huge potential of digital technologies in promoting paradigm shift (Brenner and Hartl, 2021; Bohnsack et al., al.,2021). While many argue that digitalization can be a facilitator of sustainability through efficient monitoring, analysing, and predicting (Castro et al., 2021), some worries that digitalization would be a fire accelerator as it could lead to more rebound consumption (Lenz, 2021). Existing research mainly relies on deductive approaches to speculate on the ultimate effects of digitalization on sustainability, little empirical knowledge has been developed on how digitalization impacts sustainability transitions (Anderson et al., 2021). While digitalization could foster green innovations, it may also justify the incremental 'greening' of unsustainable technologies and thus consolidate the power of incumbent actors (ibid.). More importantly, digitalization usually leads to the concentration of power to a small set of actors (e.g., digital platforms) and could result in a radical relationship change between incumbents and newcomers (Loock, 2020). Therefore, transition research should stop treating digitalization merely as a landscape factor but rather as a subject of empirical study in its own right (Sareen and Haarstad, 2021).

That said, existing research tends to view digitalization as an umbrella term without differentiating various digital technologies and the way they exert impacts (e.g., IoT as connectivity technologies, AI as automatization technologies, and blockchains as trust

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technologies) (Brenner and Hartl, 2021). This paper focuses on the role of 'trust machines' (Völter et al., 2023)—blockchain technologies— in sustainability transitions, as we notice that trust is largely taken for granted in transition research without questioning what types of trust we are advocating for and how trust matters. Blockchain allows information and transactions to be systematically recorded through a cryptographic process in a public database/platform so that all participating agents can validate the information and transactions (Schinckus, 2020). Blockchain is essentially an 'institution technology' as it can significantly lower the transaction costs of economic coordination and reform the governance between a network of economic agents (Allen et al., 2020). This decentralized characteristic of blockchain makes it well-suited to address the decarbonization, digitalization, and decentralization of many sectors, especially the energy sector (Di Silvestre et al., 2018; Andoni et al., 2019), where many incumbent and niche actors are experimenting with blockchain technologies for green solutions.

Transition scholars have begun to show interest in the role of blockchain in sustainability transitions, but the focus is primarily placed on what blockchain technology has been adopted and the related adoption barriers (Burger et al.,2016; Andoni et al., 2019; Loock, 2020; Friedman and Ormiston, 2022). Yet, we still have little empirical knowledge on why transition actors (especially the incumbents) initiate blockchain solutions and how exactly blockchain's key mechanism—trust—matters in transition practices. In particular, there exist different sources of trust, e.g., interest-based trust, personal relation-based trust, network-based trust, institution-based trust, and norm-based trust (Granovetter, 2017), with different trust mechanisms among participating agents. Moreover, trust is not necessarily always good. It may be detrimental to innovation when it exceeds a certain level (Morales et al., 2011; Bidault and Castello, 2010). Therefore, the objective of this paper is to examine the practices of blockchain-based green experiments developed by incumbent actors and niche actors, revealing how various types of trust matter in transition initiatives, and reflecting the impact of digitalization on the trust and power relationships between transition actors.

To this end, this paper designs a comparative case study on blockchain-based green experiments in China's electricity sector. Under the discourse of carbon neutrality, both incumbents and newcomers in China's electricity sector are experimenting with blockchain technologies to manage distributed energy resources (DERs). The first case is the State Grid initiating a blockchain-based renewable energy storage sharing in Qinghai, showing potential to be scaled up. The second case is about a blockchain company endeavouring to be an aggregator for the regional DER market with limited success in Shanghai. This paper compares the two blockchain applications in terms of their motivations, proceeding process, and experiment outcomes, demonstrating how trust is articulated by different actors and influences actor relations in transition initiatives. This research contributes to transition research by revealing how different types of trust matter in the interactions between transition actors and discussing the uneven power impact brought by digitalization.

This paper is organized as follows. Section 2 theorizes the role of trust and digitalization (blockchain in particular) in sustainability transitions. Section 3 provides the methodology and Section 4 compares the two cases. Section 5 discusses what types of and how trust matter in sustainability transitions as well as the merits and limitations of blockchain-based trust. Section 6 concludes.

2. Trust, digitalization, and sustainability transitions

2.1. Sources of trust

Trust is pervasive and multi-faceted, making it difficult to give a universal definition (Simpson, 2012). It can be viewed as a behaviour, a belief, an intention, an affection, or a prediction depending on the context (Mcknight and Chervay,1996). Rousseau et al. (1998:219) provide a widely cited definition of trust: "a psychological state comprising the interaction to accept vulnerability based upon positive expectations of the intentions or behaviours of another". In short, trust means one has confidence in the trusted parties that the trusted has the competencies and capacities but will not take advantage of one's vulnerabilities to behave in a way that is damaging to one, even though the trusted is given a degree of freedom to violate the expectations (Gambetta, 2000).

Trust is also conceptualized at different levels (Murphy, 2002). At the micro-level, trust is based on shared experiences or the reputation and competence of the trusted. It is relatively labour-intensive and time-consuming to earn this type of trust, but once earned, will lead to low-risk strong ties. At the meso-level, trust is based on superficial ascriptions such as race, religion, kinship, and ethnicity (Humphrey and Schmitz, 1998). This has a lot to do with historical and cultural influences and is useful when quick decisions need to be made. At the macro level, trust is based on higher-order beliefs in the formal institutions or the goodwill of society. It is more generalized and thus riskier but may enable one to bridge across communities.

Similarly, Granovetter (2017) differentiates five sources/types of trust, namely, interest-based, personal relationship-based, network-based, institution-based, and norm-based trust at different levels. Interest-based trust assumes that a rational actor engages in trusting behaviour towards another only if it is to his/her benefit to do so, that is, the expected gain from trusting behaviour is bigger than the potential loss. Here, the trustor's rational expectation is encapsulated in the interests of the trusted (e.g., to continue the relationship) —if the trusted harms the interest of the trustor, then the trusted harms his/her own as well (Hardin, 2002). Trust based on personal relationships rises from having previous exchanges with the trustee or knowing the reputation of the trustee for fulfilling expected exchanges. It is process-based confidence gained from close personal relationships, often for long periods. In this type of trust, maintaining the relationship becomes the value of interaction itself, to the extent that the transacting parties feel distressed to have cheated even if the other one did not find out. The above two types of trust mainly occur at the micro-individual level and depend on relational embeddedness.

At the meso-level, trust based on membership in groups and networks goes beyond pairwise relationships and is associated with structural embeddedness where pairwise relationships are nested in a more complex structure of social relations (Granovetter, 2017). This network-based trust is more likely to happen among those who consider themselves members of the same group (e.g., race, ethnicity, religion). The assumption here is that shared group membership will bring a common understanding and make the outcome

of the exchange more likely to be satisfactory to both parties.

As society progresses economically, trust derived from a small-scale personal relationship or group membership is insufficient to deal with the increasingly complex and large-scale social situations, where people need to exchange with people they know very little about. Here, institution-based trust (e.g., credit rating) takes up the slack and makes deception or betrayal less likely. Reliable institutions facilitate trusting behaviour by allowing individuals to develop relationships with relatively small risks and providing insurance when trust fails.

Finally, one trusts another may derive from the norms one adheres to. These norms may expect one to trust another who owes one a favour or act in a trustworthy manner in general, or trust others in one's own group. Norm-based trust is usually embedded in the culture of a nation, region, or social group, and partially explains the differences in trust levels among countries. However, as Granovetter (2017) advocates, in most real situations trusting behaviours result from a mix of these sources, and it is misleading to limit the explanation to one single source.

2.2. Role of trust in sustainability transitions

The role of trust in innovation and economic development has been highlighted in the literature. At the micro-level, trust is widely believed to facilitate learning, information change, and collaboration. At the meso- and macro-level, trust is viewed as one of the most important social capitals, a positive social good embodied in norms of networks of civic engagement and a necessary condition of social integration and economic efficiency (Putnam et al., 1992). Trust improves the quality of information exchange, especially those that are more tacit and difficult to transfer, and enhances intracommunity social ties (Murphy, 2002). Granovetter (1985) argues that trust is a cheap but critical factor for embedding social relations and ensuring future transactions. In open innovation, social trust can help to overcome information barriers and reduce opportunism during collaboration, e.g., knowledge spillover and free-rider problems (Brockman et al., 2018). Trust has a more significant impact in the context where opportunism is higher (e.g., firms in high-tech industries) and formal legal institutions are missing (ibid.). Nevertheless, too much trust may be detrimental to innovation (Morales et al., 2011; Bidault and Castello, 2010). In their investigation of a Spain industrial park, Molina-Morales et al. (2011) find that when the level of trust exceeds the optimal level, it may bring diminishing benefits and even reduce the innovation returns of the firms involved. Too much trust may lead to insufficient monitoring and cognitive lock-in, leaving firms fewer efforts and incentives to cultivate new relationships and access diverse knowledge and novel flows of alternatives.

In sustainability transitions research, the role of trust is often highlighted, but the concept of trust has been relatively under-theorized (Lawhon, 2012). Sustainability transitions involve multi-dimension and systemic transformation toward more sustainable modes of production and consumption (Markard et al., 2012). From a technological innovation system (TIS) perspective, sustainability transitions entail various processes/functions (e.g., knowledge development and diffusion, market formation, resource mobilization, legitimization) to promote the development, diffusion, and use of green innovations (Bergek et al., 2008). From a multi-level perspective (MLP), sustainability transitions require the destabilization of existing unsustainable regimes and the growth of green niche innovations under favourable landscape changes (Geels et al., 2011). In both perspectives, trust is acknowledged as an important factor in facilitating TIS functions and in empowering niche innovations to challenge incumbent regimes, but it has not been well theorized.

Existing transitions research has revealed the role in many aspects of niche innovations, e.g., the development of market (Jensen et al., 2015), the public acceptance of green technologies (Mah et al., 2021), the encouragement of green entrepreneurship (Yu and Gibbs, 2018; ŻuK and Żuk, 2023), and the management of uncertainties and risks (Bellaby, 2010). Trust is particularly related to the concept of legitimacy. Murphy (2015) argues that trust could enable new technologies, rules, and practices to be absorbed or embedded within an incumbent or alternative socio-technical regime. In some cases, people may have little interest in green technologies and projects, but strong trust in initiators (e.g., the government and regulators) could still pull the public to voluntarily support green solutions (Mah et al., 2021). At the community or regional level, acceptance of green technologies or projects could largely benefit from higher levels of trust due to place attachment and place identities (Wiersma and Devine-Wright 2014; Murphy, 2015; Yu and Gibbs, 2018).

Sustainability transitions involve multi-stakeholder engagement, but the levels of public trust in transition actors diverse significantly. Many studies have found there is a widespread public scepticism in energy companies or industries, while the science sector, NGOs, and third sector usually enjoy a much higher level of trust, be it in Philippine, Chile, or Germany (Sahakian, 2012; Vallejos-Romero et al., 2020; Häußermann et al., 2023). In the Western context, the government or public institutions are also less trusted by the public (Rayner, 2010). Especially after the 2011 Fukushima nuclear accident, there was a global widening trust gap in public institutions, which is defined as "the difference between the desirable and optimal levels of public trust and the perceived trustworthiness of governments and other trustees (most notably regulators and experts) in the context of energy transitions" (Mah et al., 2021:253). Trust gap has different dimensions, including trust in information, trust in motives, and trust in competence (ibid.). Trust in information is associated with transparency, comprehensiveness, objectivity, and reliability; trust in motives is related to openness, integrity, inclusiveness, representativeness, credible political commitment, and perceived policy outcomes; trust in competence is often influenced by capability in effective operation and capability in risk management. Therefore, a certain actor may be trusted in some dimensions but distrusted in others. For instance, Sahakian (2012) finds that in Philippine NGOs are generally trusted (in motives), but they lack the coronation to act as a unified interest group on key, systemic issues (trust in competence). Considering the sources of trust developed by Granovetter (2017), a more complete picture of trust can be depicted from two sides: sources of trust (why trust) and dimensions of trust (trust what) (Fig. 1).

From an MLP perspective, incumbent actors and niche actors in sustainability transitions face different levels and types of trust. In

the dimension of motives, niche actors are often trusted in their will to bring more sustainable solutions, while regime actors are usually believed to tend to safeguard invested interests at the cost of public well-being. However, in the dimensions of competence, niche actors often suffer from high scepticism due to the liability of newness, while incumbent actors can enjoy a higher level of trust in their capabilities due to their dominant positions in economic, political, and societal arenas. Murphy (2015) argues that the trust embedded in existing conventions, practices, motivations, and capabilities among incumbent actors could make it very difficult for niche actors to re-direct regimes change towards sustainability. In practice, it becomes more complex that many incumbent actors are also experimenting with niche innovations or adapting the regime to emerging challenges to avoid a complete breakdown (Späth et al., 2016). These incumbent actors as niche agents are typically exemplified in the automobile industry where many traditional automobile companies are entering the electric-vehicle (EV) market to adapt to the landscape change (ibid.). Previous transitions research tends to focus on how regimes are challenged by niche innovations ('impacts on regime'), Späth et al. (2016) argue future research should shed more light on how regime actors influence niche innovations ('impacts of regimes'). As trust is unevenly distributed between niche actors and incumbent actors, it is worth attention to examine how incumbent actors and niche actors compete for and leverage various types of trust to sustain their visions.

Finally, the issue of power cannot the ignored as trust is a power-laden phenomenon (Murphy, 2015).

Trust means one needs to/ is willing to depend on the other party, leaving the latter in a position of power over him/her (McKnight and Chervay,1996). The trusted party can thus exert power on the first party through dependency-based influence and control. In transition fields, Avelino and Wittmayer (2016) object to the idea that some actors (e.g., incumbent actors) are always more powerful than others, instead, they have different power relations in different scenarios. According to Avelino (2011), there exist three types of power relations: a) A has power over B, b) A has 'more/less' power than B to do x, and c) A and B have a different kind of power. Incumbent actors may have reinforcive power to strengthen existing institutions, while niche actors may develop innovative power and transformative power to develop new resources and institutions.

2.3. Digitalization and trust

When it comes to the digital era, the types of trust and their role in sustainability transitions could be redefined. Trust is a fragile resource as it is difficult and time-consuming to create but can be destroyed easily (Dovey, 2009). As digital technologies can alter actor dynamics and power relations, there are numerous ways in which digitalization undermines trust (Rijswijk et al., 2023). On the one hand, positive outcomes may be created when trust is established through the usage of digital technologies. For example, small ecological farmers in Ghana used digital technologies to close the information gap in customer service and create a data basis for credit loans (Agyekumhene et al., 2018). On the other hand, digital technologies can also lead to a decrease in trust, especially when the privacy and security of the parties involved conflicts (Connolly and Bannister, 2007). From a causal perspective, trust is a prerequisite for digital technologies to play an important role (Schmidt et al., 2020). And vice versa, digital technologies can accelerate the process of establishing trust and the speed of expanding trust networks (Fielke et al., 2020; Yan and Holtmanns, 2008).

In energy transitions, digitalization in the energy sector entails massive volumes of data created at every level of the energy value chain. It holds great promise for all segments of the energy ecosystem, including households, prosumers, distribution, transmission, generation, and retail (Küfeoglu et al., 2019). Energy digitalization is frequently related to three key technologies: artificial intelligence(AI), the Internet of Things(IoT), and blockchain (Li et al., 2022). AI-powered digitalization can boost the efficiency of energy sector, and IoT devices enable monitoring, recovery, and maintenance activities. Instead of enhancing existing energy efficiency, blockchain-enabled digitalization can aid in better network and congestion management, assisting with the renewable generation intermittency problem, allowing for more effective network monitoring and operation, and providing digital platforms for demand

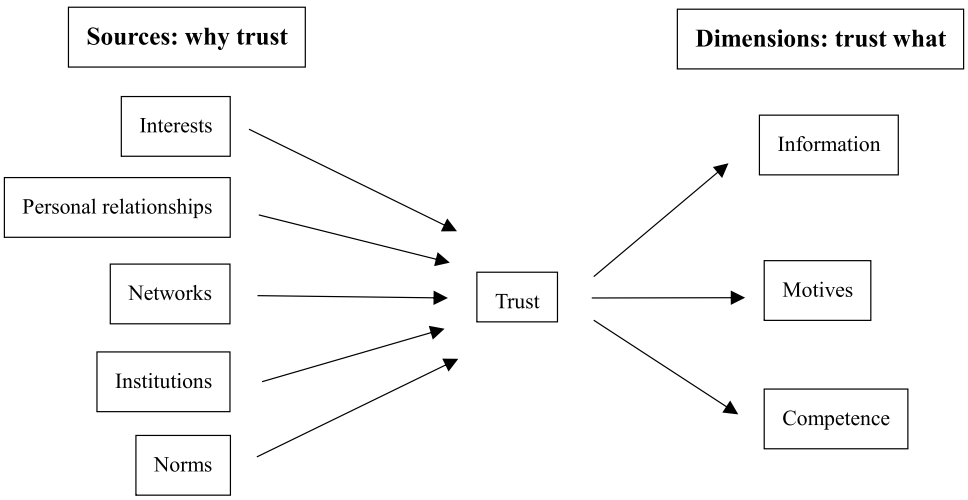

Fig. 1. A conceptualization of trust sources and dimensions.

response and peer-to-peer (P2P) energy trading (Ahl et al., 2022; Allen et al., 2020; Küfeoglu et al., 2019). While blockchain is generally believed beneficial to energy transitions, Schinckus (2020) argues that there are 'bad' and 'ugly' sides of blockchain——its huge electricity demand for computing could lead to more polluting activities and the increasing needs for computation power could bring the risk of concentration in mining industry.

Blockchains are envisioned as trust machines, referring to trust-free systems built on public and permissionless blockchains (De Filippi et al., 2020). Transactions on a blockchain network do not necessitate an established trust relationship. The system adopts a radical, decentralized accounting approach to keep an immutable, collectively agreed, and publicly visible record of past transactions and promotes a value exchange mechanism that does not require a central authority or any third-party institution (Casey and Vigna, 2018; Hawlitschek et al., 2018). People trust the blockchain network because they believe in the open-source code that can create a privacy and security network (Casey and Vigna, 2018; Shin, 2019). However, Blockchains are not all trustless system concepts. A certain level of trust is usually pre-required for users to adopt blockchain-based systems (Ahangama and Poo, 2016; Zarifis et al., 2015). The long-term viability of blockchain-based energy solutions is dependent on the establishment of transparent communication channels among stakeholders, which means that blockchain's trust-building mechanisms must take into account trusted information flow, appropriate rules, accountability systems, conflict resolution and voting mechanisms, and overall participant adaptability (Howell and Potgieter, 2019; Howell et al., 2019; Kreiss et al., 2011). Co-designing blockchain governance systems with developers, communities, and users can be a powerful method to strengthen democratic values (Valdivia and Balcell, 2022).

Overall, what/how trust matters in sustainability transitions in the digital area remains a huge knowledge gap. Previous research on the role of digitalization in sustainability transitions fails to reveal the interactive relations between trust and digitalization: trust is needed to facilitate digitalization while digitalization reshapes trust relationships between transition actors. More specifically to blockchain, existing literature has said little about how and by whom these trust machines are initiated and succeed for sustainability transitions. Adopting blockchain is not simply about adopting a certain technology, but a system of institutions and rules, which is an arena full of power struggles. To what extent can this institutional technology be adopted has much to do with who initiates and designs the system and how other agents are mobilized. This is where various sources of trust and power matter. In particular, how incumbent actors could leverage their established relationships to develop machined-based trust for sustainability purposes is largely underexplored. We therefore need to examine what types of and how trust matters in blockchain-based transition experiments, and whether and how trust relationships between incumbent actors and niche actors can be reshaped by blockchain applications.

3. Methodology

3.1. A comparative case study

This study proposed a comparative two case study about blockchain-based green experiments in China's energy sectors. Blockchain has increasingly become a key arena in China since 2019 when China's President Xi Jinping declared that China aimed to be a rule-maker on blockchain. In April 2020, The National Development and Reform Commission included "blockchain" in the scope of new infrastructure for the first time. Meanwhile, blockchain is increasingly connected to achieving China's dual carbon goals. In 2021, the National Energy Administration issued the "Guiding Opinions on Energy Work", suggesting corporates promote the digital and intelligent upgrading of the energy industry and enhance the coordinated development of distributed energy, microgrids, multi-energy complementary and other scenarios in smart cities or industrial parks, in which blockchain was particularly highlighted in distributed energy trading and clean energy consumption. National Development and Reform Commission and National Energy Administration also launched a pilot market-oriented transaction of distributed power generation, aiming to promote rooftop distributed photovoltaics and drive the deregulation of the power sales market. At the local level, various blockchain-based low-carbon projects have been initiated.

State Grid is the largest state-owned grid company in China. Realizing the huge potential of blockchain, State Grid established the State Grid Blockchain Technology (SGBT) in 2019 and then established the strategic goal of building an internationally leading energy internet enterprise with Chinese characteristics. Currently, besides having established a blockchain-based public service platform in the electricity industry –State Grid Chain, SGBT is trialling blockchain in ten application scenarios, i.e. New Energy Cloud, Power Trading, Quality Service, Integrated Energy, Material Procurement, Smart Finance, Smart Law, Data Sharing, Safety Production, and Financial Technology, in several provinces. The case we chose is about SGBT adopting blockchain to achieve shared energy storage (SES) in Qinghai. This case is completed by the interaction between the sub-chain carried by the State Grid Qinghai Company (Qinghai Grid) and the main chain of the State Grid Chain. The outcome of the case has led to certain demonstration effects in China.

Company X (anonymized) is a Shanghai-based energy blockchain start-up company, targeting Virtual Power Plants (VPPs) businesses. VPPs are the primary tool for integrating distributed energy resources into the existing electricity system at a low cost. VPP can improve the visibility and control of DERs for system operators and other market participants. Company X intended to use blockchain to reshape the relationship between industry chains in China's current electricity market. However, due to limited access to existing grid networks, the business is struggling to survive. These two cases respectively represent incumbents and newcomers in the electricity sector to initiate blockchain applications for low-carbon goals.

3.2. Data and analysis

Primary data come from semi-structured interviews and participant observations. During September 2020 and April 2022, the authors completed 14 semi-structured interviews (6 for State Grid and 8 for Company X). Semi-structured interviews were conducted

in Chinese with different stakeholders of the two practices, including both face-to-face and Zoom interviews. Participant observations create opportunities for informal interviews which can complement and help to focus further on the context of semi-structured interviews. From 2020 to 2023, the first author was often involved in blockchain-based businesses in China as an industry expert. The first author conducted participant observations when serving as a panellist, session moderator, or visiting specific industry leaders in the clean technology field and other blockchain-related conferences. In addition, the first author participated as an audience member in relevant conferences and forums to conduct random observations. Besides, second-hand materials such as government policy documents, industry reports, news, and company reports were also collected to triangulate the first-hand data. The data analysis process is based on an interpretivist viewpoint, which includes understanding the world through interactions with others, realizing our dynamic relationship with the data, and embracing our role within the research (Greenbank, 2003). The authors use an open coding strategy to build a participant-generated 'theory' from the data (Strauss and Corbin, 1998). As emergent codes, open coding entails using codes obtained from the text. We code conceptually related events/actions/interactions in this study, emphasizing that we code concepts rather than lines of data (Blair, 2015).

4. Comparing two blockchain-based green experiments in China

4.1. SES by State Grid in Qinghai

Qinghai is a less developed region in northwest China but shows remarkable development in renewable energy. By 2021, Qinghai's installed capacity of renewable energy reached 24.85 million kW, accounting for 61 % of Qinghai's total capacity and ranking first in China. However, Qinghai's renewable energy market mostly relies on the developed provinces in East China and faces enormous pressure of wind and solar energy curtailment. "*At the worst time, the curtailment rate was basically 50 %*" (Interview, Grid Expert). Moreover, due to its high instability, electricity generated from wind or solar was often called '*garbage electricity*' within the grid system (Interview, SGBT marketing manager). Though the curtailment rate of renewable energy was largely reduced in the late 2010s mainly because of the construction of extra-high voltage transmission channels, the local branch of State Grid, Qinghai Grid, faced great political pressures to meet the government's assessment of two indicators for renewable energy: curtailment rate and renewable energy consumption capacity.

Therefore, Qinghai Grid looked for innovative experiments to meet these assessments and came out with the idea of SES in 2018, with the intellectual support from North China Electric Power University. The birth of this idea was based on solid investigations on local renewable energy power plants, which also had built many electricity storage capacities required by government regulations but lacked the willingness to use the storage facilities as the cost of energy storage was very high before 2018. The grid company did not want to pay for energy storage, and neither did the local government, making it a '*dead end*' (Interview, Grid Expert).

Things changed when the cost of energy storage was significantly reduced to RMB 0.85 per kW/h, while the feed-in tariff for the renewable energy power plants was between RMB 1.05 to RMB 1.15. Thus, there existed a price difference between RMB 0.2 and RMB 0.25 per kW/h. Though not big, "*(the figure) is at least positive, not negative, so the only question is how to divide this profit*" (Interview, Grid Expert). The model of SES was thus put forward, including two mechanisms: a) Market-based trading: power plants and energy storage stations are traded through bilateral negotiations and market bidding to reach trading agreements that include trading period, scale, and price; b) Direct allocation from the grid: when market-based trading is not reached, the grid will directly use energy storage at an agreed price, releasing it when the grid has space to accept it. Both Qinghai Grid and local Energy Bureaus believed this model would be feasible, because "*the grid company does not need to pay, and the government does not need to subsidize, and power plants could still make profits… so basically it was a win-win situation*" (Interview, Grid expert). This created an interest-based trust among the participants.

The key to this model is to reach information symmetry among renewable energy power plants, grid company, and energy storage stations regarding e.g., who store and use energy at what scale and on what prices for financial settlement, which was the biggest concern for energy storage stations. At that time, the normal settlement process was that the grid company settled with the renewable energy power plants, and then the latter settled with the energy storage stations. Although the initial cooperation was with state-owned renewable energy power plants, energy storage stations worried that "*the settlement cycle and the reliability of the settlement were not completely controllable, as there were many problems in the middle that could not be controlled, resulting in their revenue not being paid back as expected*" (Interview, Grid Expert). To address this, the SES model proposed a direct payment scheme between the three parties. For instance, if one kWh stored energy were released to the grid, then the RMB 0.25 revenue would be directly shared between energy storage stations (80 %) and renewable energy power plants (20 %). However, as the model proceeded, increasing needs were developed for the speed and simplicity of settlement, and here was when blockchain stepped in. As a grid expert explained:

> *"Why was blockchain introduced then? The problem is the speed of settlement, or the simplicity of settlement, and blockchain solves this problem. So, when we considered the introduction of blockchain, it was mainly because of the symmetry of data and the credibility of the settlement". (Interview, Grid Expert)*

Meanwhile, as blockchain technologies become more mature, SGBT was looking for blockchain application experiments. Qinghai's shared energy model was selected as one of the pilot cases and blockchain technologies were introduced as the backbone of this SES model. As State Grid lacked endogenous technical knowledge on blockchain, it outsourced the construction of the blockchain infrastructure to a blockchain company, who encountered many difficulties to understand the grid system. As a technical manager illustrated:

"The difficulty we encountered is the understanding of the traditional grid system, in terms of either technology or process model, the understanding of these aspects is a large difficulty…. We spent two years and a lot of energy to understand the electricity industry, this is also the most difficult part for us now". (Interview, blockchain company manager)

Similarly, the grid system also had to learn the idea of blockchain. As a SGBT marketing manager illustrated: "*when we introduced blockchain technologies, we did special promotions of blockchain for everyone, and only after they had more special knowledge on blockchain, did we introduce this concept, and we had done a lot of paving*" (Interview, SGBT marketing manager). Qinghai Grid followed the intention and forms an innovative market operation model by setting up bilateral negotiations and market bidding on the chain (Fig. 2).

The blockchain applications enabled the SES model by incentivizing the energy storage stations and energy storage facilities to participate in peak shaving auxiliary services to gain profit. For Qinghai Grid, though no instant profits were generated for it in this blockchain-based model, it had some more strategic considerations. As a marketing manager illustrated:

"For the State Grid, building blockchain-based energy storage is kind of our auxiliary services, from which we did not earn money. We just provided a public platform for everyone. Though we do not earn money now, in the future, we may charge some fee if more participants move in. Also, from the beginning, State Grid did this is to increase the voice among the state-owned (energy) enterprises. State Grid itself also has to bear the responsibility of the consumption of renewable energies, so we invested in this project of tens of millions of Renminbi, which is quite a lot." (Interview, SGBT marketing manager)

Informants from the State Grid did not deny its ambition to dominate the blockchain platform. For State Grid, it didn't have the problem of information asymmetry since it had access to all the trading data along the value chain. However, State Grid could gain more influence along the value chain through facilitating trust between the energy generators and energy storage stations. Its dominant position in linking the energy generation side and storage side making it the sole candidate with the capacity to build the blockchain-based SES platform. Naturally, State Grid became the leader and hub operator of the blockchain alliance, allowing it to have more say in setting the rules of the blockchain platform (Interview, blockchain company manager). Indeed, its dominant position in electricity transmission made it relatively easy to gain personal relationship-based trust and network-based trust and thus mobilize other stakeholders to join the platform. In Qinghai, State Grid managed to ally two state-owned power generation enterprises, which owned hundreds of wind and solar power plants and energy storage stations. For power plants and energy storage stations, they also hoped to symmetry the information flow with the grid side. With the blockchain platform applied, "*all the power plants and energy storage stations involved can see how many transactions they have made with the grid in real time*" (Interview, blockchain company manager). In the past, data verifications were mainly based on the Grid company's data in a centralized way. Through blockchain, the system enables a transparent multi-agent interaction through transparent dispatching and transparent trading. Transparent dispatching means that all participants can see the source and use of electricity and transparent trading enables the costs generated by user transactions to be allocated directly through smart contracts.

However, the main hurdle was that whether the participants were willing to share their relevant data on the platform. Some renewable energy power plants were reluctant to join the platform because they had stable customers and most of their electricity could be consumed directly. For others, the key concern was being too dependent on State Grid. As an informant said:

"They (the participants) showed much enthusiasm to join, but they also worried that State Grid might stuck their necks…they are afraid of over-reliance on our platform…the key concern is, all the data go to the State Grid." (Interview, SGBT marketing manager)

In other words, not all the stakeholders were willing to be totally transparent and over reliant on a dominant player. Even within the grid side, there were concerns regarding interest conflicts after being too transparent on the blockchain. As a blockchain company manager explained:

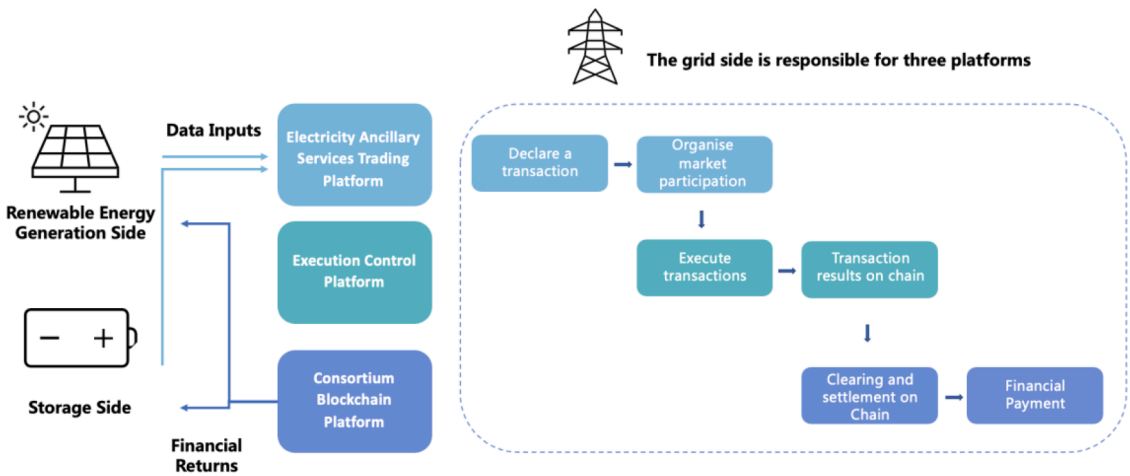

Fig. 2. The SES platform operation flow.

"When all the data become consistent and transparent, the previous interest chain within the grid system may be affected. In the old system, data were not transparent, it was a black box, and there were many grey areas, which could become the resistance of SES model promotion." (Interview, blockchain company manager)

A university expert who was the consultants of the SES model also expressed similar observations: "*we spent more than one year to promote information disclosure, but this process was very difficult. The grid company is too powerful, it has all the key information, and it's hard to push for some information when it doesn't want to disclose it*" (Interview, Grid Expert). Similarly, some power plants and energy storage stations preferred to have some 'grey space' despite of financial loss. "Data security is an issue, being dominated is another issue" (Interview, SGBT marketing manager).

Despite these concerns, the blockchain-based SES model showed obvious advantages in grid balancing, demand responses, alleviation of electricity fluctuations in renewable energy generation, and fully released energy storage capacities. It helped to reduce renewable energy curtailment, improve the grid's ability to absorb more distributed energy resources, and effectively improve the safe and stable operation of the grid system. By August 2021, the blockchain-enabled SES pilot had accomplished a total of 2509 transactions for peak-shaving auxiliary services, with charging and discharging efficiency of 80.78 %, and 543 h of annual utilization. 347 renewable energy power plants had participated in the SES peak-shaving auxiliary services transactions, which accumulatively increased power generation by 68.15 GW. This relatively successful model has begun to induce a demonstration effect in China. After Qinghai, other provinces such as Hunan and Shandong have also proposed similar operation models.

4.2. VPPs by Company X in Shanghai

VPPs are the primary tool for integrating distributed energy resources into the existing electricity system at a low cost. Shanghai was included in the second batch of pilot areas for distributed electricity by China's Energy Bureau in 2016, which created a friendly policy environment to the growth of VPP businesses. Currently, the VPP pilots in Shanghai are primarily demonstration projects for demand-side management of commercial buildings, with the main participants being some commercial and industrial loads and energy storage (Dorfleitner et al., 2021). However, the policy implementation and project implementation of the currently built VPP platform is uneven, with no unified standards or interfaces. The market is mainly fragmented with different market players building their own VPPs.

In 2017, Company X started its VPP business in Shanghai. Before Company X, the founding team was the core team member of X Labs, which was one of the first companies in the world to explore the energy blockchain. The founding team developed expertise in introducing the latest blockchain technology for experimentation in countries such as Singapore, Philippine, India, Japan, Thailand, and the Netherlands, where the markets are open for regionally DERs. Due to the covid-19, Company X's main business shifted to the domestic Chinese market because of the inconvenience of communicating with overseas partners, and aimed to leverage blockchain as an aggregator role for DERs in China. Through VPP, the company aimed to participate in the integrated operation of the large number of new distributed energy assets required in the carbon neutral transition process, involving generation, storage, and load side (Fig. 3). In the company's planning, a large number of small and medium-sized distributed resources would be managed, coordinated, and optimized in a unified manner using decentralized computing and dispatching based on blockchain technology, in conjunction with AIoT terminals to achieve metering and communication.

In Shanghai, the company was certified as a load aggregator. In comparison to other markets where variable load is the norm, Shanghai's load policy progress has been slower. Despite the addition of some adjustable load capacity, the number of access equipment was very small. When Company X tried to introduce their blockchain application to their customers in Shanghai, it encountered many difficulties. In the following, we illustrate Company X's attempt to apply blockchain on the load side.

Referencing its previous overseas experience, Company X pre-defined two types of user groups in Shanghai: household users and commercial users. Like most blockchain ventures, Company X employed an open-source approach to attract clients and build trust. However, unlike other blockchain projects where the primary business models and interaction mechanisms are all carried out on the blockchain, energy blockchain projects must manage the trust relationship that exists between on-chain and off-chain data. The Oracle system developed by Company X promised to connect blockchain and off-chain electrical devices, ensuring the safety and accuracy of off-chain power usage and transaction data, and making them blockchain-compatible. Instead of encrypting all data in the operator's system, the Oracle allows the operator to simply be responsible for the deployment of the equipment. Multiple nodes in the blockchain complete the entire data measurement and billing procedure. Users can always double-check the measurement data on the comparison chain and their own electricity use.

However, participating in electricity trading is a complex process that requires an understanding of the electricity price curve to change trading strategies at any time. Most household and commercial users had no idea about energy trading strategies. The lack of public knowledge on VPP technology and business constituted the first barrier for Company X to gain public trust and acceptance.

"The key lies in the understanding of this business, which is quite new in China. When we talked to some factories and introduced VPP businesses, they didn't have much sense and asked what is VPP?" (Interview, Peer Company CEO).

Under such a context, Company X aimed to be a virtual agency to set long-term and automated strategies for household users.

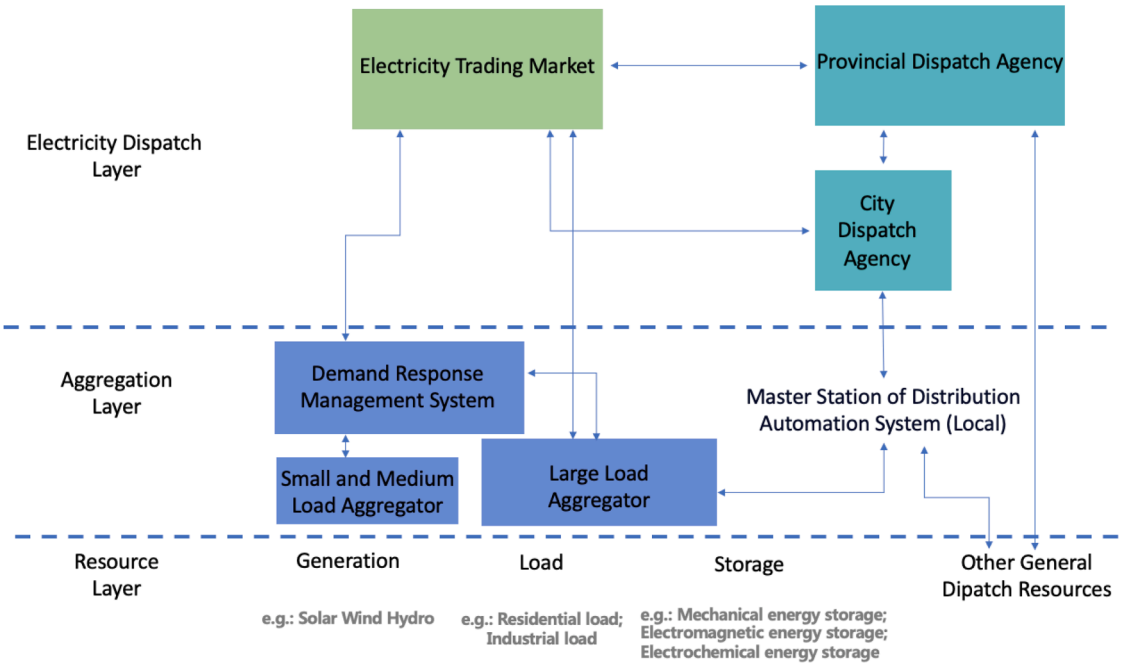

Fig. 3. Company X's Vision in Chinese Electricity Dispatching System.

Company X utilized blockchain to provide modular services for the asset owners, such as aggregator-as-a-service to the owner's energy storage, charging points, or building HVAC.[1] This strategy turned out not successful. "*In 2019, the adjusted load was only a few megawatts, which was insufficient*" (Interview, Company X director). The key problem is that Company X's founding team were all from blockchain-related backgrounds, and they lacked an in-depth understanding of the Chinese electricity market. Company X also ignored the huge differences in the electricity market in terms of institutions and infrastructure between China and Southeast Asia countries, where the company had previously experimented with electricity trading. Unlike in the Philippine or Thailand, Chinese household users are not participants in the electricity market due to the lack of openness of the retail electricity market. On the one hand, the current grid infrastructure cannot afford all Chinese in the energy trading market. On the other hand, household users in China enjoy a low tariff subsidised by the State Treasury (Lin and Jiang, 2011). Residents buy electricity from state-owned grid companies rather than directly from power companies. For Company X, the profit of its service was too thin (Interview, Company X director), and for householder users, the financial gains were also barely visible, resulting in a low interest-based trust.

> *"The electricity market for consumers has never been a market-based mechanism but rather a livelihood project, with residential electricity prices in China averaging around RMB 0.50/kWh." (Interview, Electricity market analyst).*

In the commercial scenario, the situation was slightly healthier. Company X targeted commercial users such as real estate developers or industrial parks, who were usually equipped with DERs rather than centralised power stations. In order to have a green premium or image, these asset owners needed to have a third-party certification to prove that the electricity they generate or procure is green. Besides, in order to achieve aggregated distributed energy and participate in the peak reduction and valley-filling business of the electricity market, the industry chain must be digitalized for the overall deployment of resources. Company X identified this value gap and aimed to utilise blockchain to help the DER owners to achieve asset digitalization and financialization. It used blockchain's digital signature, consensus mechanism, and smart contract technologies to build a digital ID system for the corroboration of distributed energy assets. For example, Company X considered that DER data on the blockchain can be consolidated into pools for load management, which invokes distributed energy for load allocation through market mechanisms.

> *"Distributed energy assets with small ownership rights can be allocated, coordinated, and traded in a unified manner after DER asset data on the blockchain… although distributed power trading will reassign the vested interest from the grid side, as a load aggregator, it helps the grid to manage loads and reduce management costs," (Interview, Company X Senior executive).*

However, Company X encountered many challenges in entering the electricity market, where the inherent interest pattern of the electricity industry is solid. As an outsider of the energy system, the lack of established relationships and trust in the electricity sector has made Company X very difficult to persuade its customers to use blockchain and access to the dispatch of distributed energy storage

[1] The use of various technologies to control the temperature, humidity, and purity of the air in an enclosed space is referred to as heating, ventilation, and air conditioning (HVAC).

and charging points. As a director of Company X explained:

> *"To join the market, we must first gain asset owner acceptance and sell our product such as SaaS[2] to large asset owners (e.g., energy storage operators). However, we are unable to sign contracts with them and hence cannot access the mainstream market. It necessitates a great deal of guanxi (relationship), particularly in China…It is similar to engineering outsourcing, it takes a lot of effort because you have to build relationships" (Interview, Company X director).*

The team members of Company X only realized this problem later that the blockchain solutions strengthening the electrical market are largely different from solely blockchain applications.

> *"The electrical market is a bit special. The electrical market faces particular difficulties, including how to distribute resources among multiple industrial giants… Because you are not from this industry, it will be tough for you to tackle industry-specific challenges." (Interview, Company X director).*

Meanwhile, adopting blockchain in the electricity value chain is not simply about incorporating an added-on technology to the existing system. Instead, it entails system-level redesign in terms of rules and infrastructures. This formed another big barrier for the electricity sector to accept blockchain. In addition, there were many trust concerns about how the blockchain-based solutions would impact the original system. As a director of Company X observed:

> *"When there is no blockchain in the original business process, it runs smoothly; however, if you add a blockchain suddenly, the original system will be broken, resulting in some uncontrollable situations. [The customers worry] What should you do if something goes wrong with your blockchain?…… you don't know whether the blockchain-based system is reliable or not, no one will risk running a new system, especially when there are only a few economic returns" (Interview, Company X director).*

Another problem was that the electricity dispatching system had not been fully digitised. The electricity dispatching system was dominated by a manual dispatching system, with phone calls as the main way to communicate transactions. The electricity dispatching system was less responsive to micro, digitised distributed energy assets, which were often rejected by the dispatching system for the reasons of grid stability. Thus, for grid companies, the operational support for such aggregation services provided by Company X was instead a burden. A director of Company X believed that the company's strategy was problematic from the beginning:

> *"The concept of a virtual power plant is still not ready for us. As a startup company, we must first decide how to make money……Be big enough first, then we can help them perform such power trading and use some of the redundant energy to negotiate and create a deal with the power grid. These medium- to long-term transactions will incur some processing fees for us." (Interview, Company X director).*

Though Company X intended to establish a mechanism based on open-source blockchain technology by disclosing the code and ensuring data flow transparency through Oracle, it currently lacked the ability to connect all related parties due to power and knowledge asymmetry. Interviewees suggested that blockchain seemed more suitable for incumbent giants to redivide the pattern of interests… because they are "*more central, more powerful, and then they can use blockchain to redraw the pattern of interests. However, pure blockchain technology [providers] cannot undermine the underlying trust and interest patterns*" (Interview, Company X director).

As a result of these factors, the growth of this commercial business is still far behind the company's planning. So far, the connected assets are less than 10 MW. Company X has only formed alliances with certain factory loads and building HVAC. The grid companies prioritized demand-side responses that are characterized by human decision-making and complex processes that do not allow for real-time load response. This scale is far from the prospect of distributed clearing and settlement of DERs enabled by blockchain. An electricity market analyst provided some reflections on the causes:

> *" From the grid's perspective, the grid is currently doing its best to avoid this piecemeal approach to dispatch… VPPs for dispatching necessitate forcible control of the equipment, and establishing a VPP mechanism without a sufficient foundation of trust is impossible. At this point, company X has not achieved the level of forcible control over the equipment" (Interview, Electricity market analyst).*

Off-grid customers were reluctant to give control of their equipment to Company X, and therefore could not provide a stable demand-side response to the grid. Without sufficient trust relations with other participants, Company X failed to establish stable institution-based trust on its own, preventing it from connecting with more distributed energy owners. Currently, the company is standing at a crossroads. Company X hopes to expand its business in the future by partnering with or being acquired by large energy storage companies.

5. Discussions

5.1. Case comparison

It can be seen that the outcomes of the two cases are attributed to multiple factors, e.g., actor position in the market, knowledge about technologies, and economic returns. The types and levels of trust are embedded and reflected in these factors. Table 1 compares the main features of the two cases. SES in Qinghai was led by a powerful incumbent while VPP in Shanghai was initiated by a

[2] Software as a service (SaaS) allows users to connect to and use cloud-based apps over the Internet.

newcomer to the electricity sector, demonstrating significantly different motivations, processes, and outcomes. For State Grid, it has a strategic vision to use blockchain to regain its dominant influence in China's electricity value Chain. Since China's grid market reform in 2002, State Grid was gradually 'channelized' as a grid transmission company, with upstream and downstream businesses being 'cut off' (Interview, Grid Expert). Blockchain caters to its vision by exerting its trading and data authentication attributes in the electricity sector. The SES model in Qinghai was seen as an experiment to fulfil the political mission of reducing renewable energy curtailment rate on the one hand and regaining dominant influence along the value chain. Thanks to its incumbent position and established trust relationship, State Grid managed to introduce the SES model in Qinghai with a relatively successful outcome, which is expected to be scaled up. Blockchain solves the symmetry of data between multiple entities and the credibility of settlement. This process entails the incumbent's established interest-based trust, personal relationship-based trust, and network-based trust to promote the institution-based trust system. State Grid managed to confirm a hub position in this model, but this also makes some participants worry about being too dependent on State Grid. The model also discourages those who rely on opaque data to gain values. From the perspective of trust dimensions, State Grid gained trust mainly in competencies, but still less in motives and information.

In contrast, Company X's motivation is to build a blockchain-based distributed energy trading platform with machine trust that is parallel to the existing system. Strategically, Company X's primary focus was on the development of blockchain technology at the expense of understanding the Chinese electricity sector. Due to the lack of industry knowledge and established relations, Company X had difficulty in gaining the trust, in particular, interest-based trust, network-based trust, and personal relationship-based trust, from incumbent players. Despite the Company's motive being understandably trusted, it fails to earn enough trust in their competence. The low level of trust is attributed to not only the company's weak market position but also the market's lack of understanding and risk concerns about blockchain-based systems.

In both cases, the blockchain-based models combine the characteristics of data governance and financial governance. State Grid uses blockchain technology to master the data of upstream and downstream industries. While improving value chain transparency, it promotes interest-based trust by assisting energy storage and power generators in optimising the clearing and settlement process. In the contrast, Company X failed to cross institutional hurdles and directly bring economic benefits to users. There are situations where traditional rules and on-chain rules support the same event in parallel, which increases the cost of trust. It demonstrates the important role of prior experiences in establishing good expectations that lead to trust. This is in line with Rousseau's et al. (1998) argument that when there is little prior experience and the relationship is relatively new, actors are inclined to use calculation (interest-based) trust based on clear rules and codes.

5.2. Blockchain, trust, and power

Blockchain is viewed as a promising candidate technology in responding to the decentralization, digitalization, and decarbonization trend of the energy sector (Amenta et al., 2021; Di Silvestre et al., 2018). A key characteristic differing blockchain from other technologies is that blockchain is an institutional technology (Allen et al., 2020). That is, the primary advantage of blockchain lies in lowering verification costs and governance costs between a network of economic agents, rather than in promoting the productivity of individual economic agents. Blockchain reduces system-level transaction costs by providing a public good, trust, which is traditionally often produced by specific intermediaries (Amenta et al., 2021). Based on a set of protocols and code-based rules developed through formal programming languages, blockchain provides a platform-wide institution-based trust that may make actors feel confident enough to engage directly in peer-to-peer markets or regulated markets dominated by some monopolies (ibid.). By providing a system-level trust, blockchain reduces the amount of trust needed in these hub operators and avoids the time-consuming trust building processes at the individual level. This system-level trust is guaranteed through predicable principles and rules that are not vulnerable to actors' personal reliability. Hence, Lumineau et al. (2021) propose the idea of blockchain governance to complement the traditional contractual governance and relational governance, which are respectively enforced by formal legal authorities and informal social norms. Blockchain governance is a self-contained and autonomous systems of rules. It entails high design cost but can reduce searching, monitoring, and enforcement costs through codifiability and verifiability.

Rebuilding trust relationships may entail rebuilding the interest system as well. In the SES case, the State Grid is the hub firm that has two sided relationships with renewable energy producers and energy storages, giving it the advantage to exploit the knowledge asymmetry over any other participants. The hub operator created an environment of trust among all sides. The relative success of this SES model is partly attributed to State Grid's dominant position in the value chain and established trust in various forms, especially interested-based trust, and personal relationship-based trust through frequent business interactions. State Gird, as a state-owned incumbent giant in the grid industry, can easily earn trust in its competencies. Though some participants were somewhat forced to join the SES platform, the visible revenue from storage profit distribution also brought them interest-based trust. This is in line with Skardon's (2011) argument that the hub firm's trust relationship must extend to external social and professional networks to recruit participants and technology providers. Therefore, in the blockchain businesses, energy incumbents may enjoy more trust and power to promote this institutional technology. Though decentralization is one of the key features of blockchain, State Grid designed its blockchain platform with many centralized rights considerations in mind and managed to centralize the data flow of the generation side, grid side, load side, and storage side on the SES platform. Therefore, this decentralization tool in fact may be leveraged to centralize power by powerful incumbents.

However, gaining such a dominant influence also brings some negative concerns. As blockchain enforces system-wide transparency and accountability, those who benefit from information asymmetry may protect the status quo and against blockchain applications (Friedman and Ormiston, 2022). As shown in the SES case, many participants, including the State Grid, were the beneficiaries of the non-transparent status quo. Not all the incumbents were willing to accept this 'forced trust' by blockchain. Platform participants also

Table 1
A comparison between the SES and VPP cases.

	SES by State Grid	VPP by Company X
Hub operator	Incumbent	Newcomer
Motivation	Political mission and maintaining a dominant position in the grid value chain	Building a blockchain-based distributed energy trading platform with machine trust
Main participants	Renewable energy generator and storage owner	Commercial and industrial DER users
Outcome	Pilot project with demonstration effect	Very little success
Sources of trust	Institution-based trust, personal relationship-based trust, interest-based trust, and network-based trust	Institution-based trust
Trust dimensions	More trust in its competence, but less in motives and transparency	Few trust in competence, relatively more in motives
Barriers	Participants' fear of being too transparent and too dependent on the hub operator	Lack of industrial knowledge and established trust relationship

worried about being too dependent on the hub operator. When the hub operators become increasingly powerful, participants' distrust in their motives and information may rise as well.

In contrast, Company X barely survived in the cracks of the incumbent grid system due to many forms of resistance. The company lacks knowledge and relationships in the grid market, not to mention various trusts. As an emerging technology, the complexity of blockchain technologies forms a functional barrier (Friedman and Ormiston, 2022) for incumbent DER players to accept them. In SES's case, it is the incumbent State Grid proactively outsourced the technology to a blockchain company that helps to earn the company some trust among incumbents. Even with the endorsement of the powerful incumbent, the blockchain company still took years to understand the electricity sector and get the technology understood in the SES value chain. Moreover, without any established relationship with DER players, Company X suffered from a severe lack of interest-based trust, personal relationship-based trust, and network-based trust to promote the complex trust system among DER actors.

In the digitalization era, incumbents may be more capable of defending their positions through the application of digital technologies in incremental innovations (Loock, 2020). Many digital technologies (e.g., blockchain and IoT) inherently involve system-level collaboration and coordination between various existing agents, which is where the strength of the incumbent lies. A powerful position of incumbents can facilitate the development of different types of trust and hence the adoption of digital applications along their value chains, which in turn strengthens their power position. In contrast, newcomers in digitalization lack initial trust and power to enter the incumbent system, not to mention bringing system-level transformational changes.

6. Conclusions

Linking digitalization with sustainability transitions, this paper explores how blockchain technologies matter in sustainability transitions through its key mechanism-trust-by comparing two blockchain-based green experiments in China's electricity sector. The findings show that blockchain technologies can bring institution-based trust among participating agents. However, blockchain-based green transitions need various types of trust to promote the institution-based trust to a system level. These born characteristics of blockchain technologies bring different meanings to incumbents and newcomers. For newcomers, due to the lack of other types of trust, blockchain start-ups face many challenges in development and scale-up in the energy sector. For incumbents, they have the inherent advantage of mobilizing established trust relationships to promote blockchain-based green experiments and strengthen their power positions. Blockchain applications are expected to bring a decentralized structure but it could end up in a centralized structure with the incumbent hub incumbent having most of the power. What we argue is that trust relationships would not be automatically reached by simply adopting blockchain technologies, instead, it largely depends on who initiates the system and whether they have existing trust relationships to establish new trust relationships, which in turn reshape existing power relations.

It should be noted that the two cases examined in this study are in China's energy sector, where the market is not fully market-oriented and state-owned enterprises play a dominant role. Whereas in many countries the energy market is market-oriented and energy companies are less trusted (e.g., Vallejos-Romero et al., 2020; Häußermann et al., 2023), state-owned energy enterprises in China usually enjoy higher levels of trust by the public as they are not fully profit-seeking but also deliver social values. This context-specific market structure and trust characteristics may exert different power relations among transition actors in different sectors or countries. Future research could further study the role of digitalization in sustainability transitions in different sectors and countries, and explore how different types of trust work.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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